

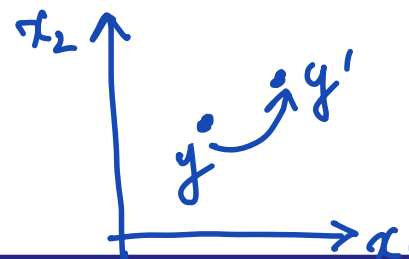
Week 7 Recitation

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Monotonicity and Nonsatiation

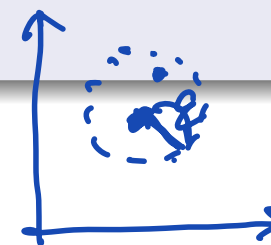


Definition (Monotonicity)

Preference $\succeq$ is monotone (i.e., nondecreasing) if for $\forall y, \bar{y} \in \mathbb{R}_+^N$ such that $\bar{y} \geq y$, then $\bar{y} \succeq y$.

Definition ((Local) Non-Satiation)

Preference $\succeq$ is Locally Non-Satiated if for $\forall y \in \mathbb{R}_+^N$, $\forall \varepsilon \in \mathbb{R}_{++}$, there exists $\bar{y}$ s.t. $\|y - \bar{y}\| \leq \varepsilon$ with $\bar{y} \succeq y$.



Monotonicity

Direction ✓



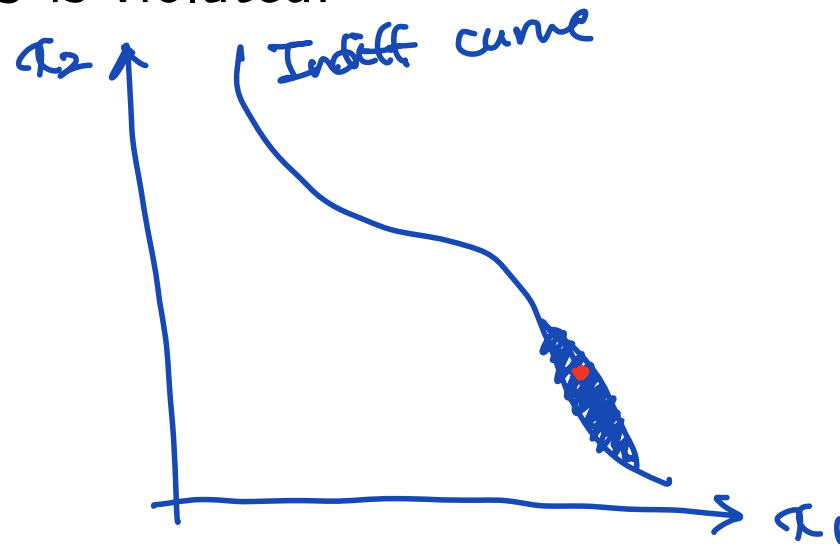
Local Non-satiation

existence

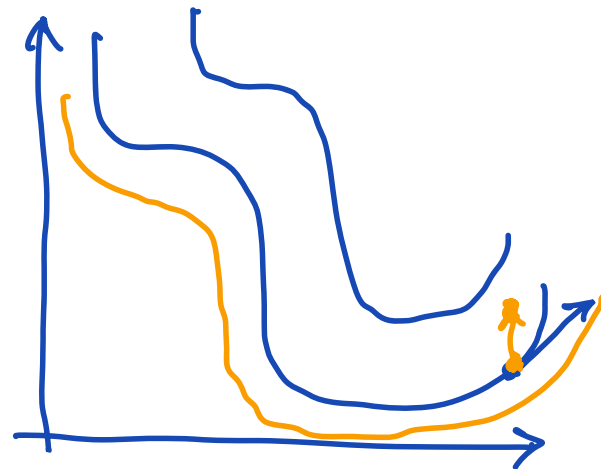
Monotonicity and Nonsatiation

Examples:

- 1 When LNS is violated:



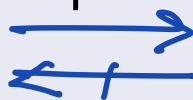
- 2 When LNS holds, but not monotone:



Monotonicity and Nonsatiation

Proposition

Monotonicity of a preference implies the Non-satiability of the preference.

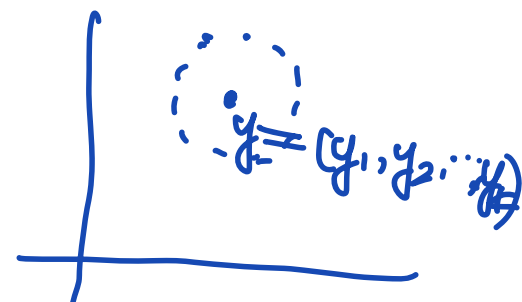


Proof.

Suppose $\succeq$ is monotone.

Take any $y \in \mathbb{R}_+^N$ and $0 < \varepsilon < 1$.

Set $\bar{y} = (y_1 + \frac{\varepsilon}{\sqrt{K}}, y_2 + \frac{\varepsilon}{\sqrt{K}}, \dots, y_K + \frac{\varepsilon}{\sqrt{K}})$



$$\|\bar{y} - y\| = K \cdot \frac{\varepsilon^2}{K} = \varepsilon^2 \leq \varepsilon$$

So we can see the LNS

Constraint and Non-satiation

Proposition

If the preference $\succeq$ satisfies Local Non-Satiation and u represents $\succeq$, then the constraint binds at the optimum.

Convexity of preference & its representation

We say $\underbrace{u}$ represents the preference $\underbrace{\succeq}$ when

$$x \succeq y \iff \underbrace{u(x)} \geq u(y)$$

Definition (Convexity of preference)

Let the bundles $y, \bar{y} \in \mathbb{R}_+^N$ where $y \succeq \bar{y}$. The preference $\succeq$ is convex if for $\forall \lambda \in [0, 1]$, $\lambda y + (1 - \lambda)\bar{y} \succeq \bar{y}$.

Definition (Quasi-concavity)

A function $f : \mathbb{R}^N \rightarrow \mathbb{R}$ is quasi-concave when for $\forall \lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda)y) \geq \min\{f(x), f(y)\}$$

Convexity of preference & its representation

Proposition

Suppose that u represents the preference $\succeq$.

u is quasi-concave iff the preference $\succeq$ is convex.

Proof.

$$x \succeq y \Leftrightarrow u(x) \geq u(y)$$

$$\lambda \in [0, 1]$$
$$\lambda x + (1-\lambda)y \succeq y$$

$(\Rightarrow)$ Suppose u is Q-concave.

Fix any $x, y \in \mathbb{R}_+^N$. WLOG, assume $u(x) \geq u(y)$.

$$u(\lambda x + (1-\lambda)y) \geq u(y)$$

$$\lambda x + (1-\lambda)y \succeq y$$

$$(\Leftarrow) \quad \begin{array}{l} x \succeq y \\ \lambda x + (1-\lambda)y \succeq y \\ u(\lambda x + (1-\lambda)y) \geq u(y), \quad u(x) \geq u(y) \end{array}$$

<Qual Sample 6>

Name:

1. [50 points] Consider an individual choosing continuous amounts of the goods $y = (y_1, y_2) \in \mathbb{R}_+^2$ who maximizes the preference function

$$f(y) = 2 \log(y_1) + 3 \log(y_2).$$

The individual cannot purchase y_1 and y_2 directly, but instead obtains them by purchasing $x = (x_1, x_2, x_3) \in \mathbb{R}_+^3$ and using a “production process.” We have production processes given by

$$y_1 \leq \min\{2x_1, 3x_2\};$$

$$y_2 = 3x_2 + x_3.$$

Purchases of x are constrained by a linear budget constraint $p_1x_1 + p_2x_2 \leq m$ where prices are $p = (p_1, p_2, p_3) \in \mathbb{R}_{++}^3$ and money is $m \in \mathbb{R}_{++}$.

- (a) Are the preferences given by $f(y)$ convex? Justify your answer. [5 points]
- (b) Do the constraints from the production processes give a convex set of (y, x) ? Justify your answer. [10 points]
- (c) Does the budget constraint on x define a convex set? Justify your answer. [5 points]
- (d) Set up the preference maximization problem. [10 points]
- (e) Is this a convex optimization problem? Justify your answer. [5 points]
- (f) Without solving the problem, can you say whether there are regions where some x will never be purchased? Justify your answer. [5 points]
- (g) Without solving the problem, can you say whether the budget constraint will bind? Justify your answer. [5 points]
- (h) Normally, when there are two goods and a linear budget, the goods are required to be substitutes. Do you think this intuition holds for the goods

y when there is a production process generating the final goods? Justify your answer. [5 points]

2. [50 points] Suppose an individual is choosing continuous quantities of goods given by $y = (y_1, \dots, y_N) \in \mathbb{R}_+^N$. Suppose the individual faces a linear budget constraint with observable prices $p \in \mathbb{R}_{++}^N$ and money $m \in \mathbb{R}_{++}$ so that goods satisfy $\sum_{n=1}^N p_n y_n \leq m$. Suppose the individual wants to maximize preferences represented by

$$f(y) = y_1 + \sum_{n \neq 1} \alpha_n \log(y_n)$$

where $\alpha_n \in \mathbb{R}_{++}$.

- (a) Is this a concave maximization problem? Justify your answer. [5 points]
- (b) Write down the Lagrangian for the above preference maximization problem. Be sure to include all relevant Lagrange multipliers. Justify any simplifications. [10 points]
- (c) Write down the first order and complementary slackness conditions for the Lagrangian in (b). [10 points]
- (d) Let $V(p, m, \alpha)$ be the value function for the preference maximization problem. Using the first order conditions and complementary slackness conditions, find the expression for $\frac{\partial V}{\partial p_1}$. You can use the envelope theorem to check your answer, but not to derive the relation. [10 points]
- (e) Let $V(p, m, \alpha)$ be the value function for the preference maximization problem. Using the first order conditions and complementary slackness conditions, find the expression for $\frac{\partial V}{\partial m}$. You can use the envelope theorem to check your answer, but not to derive the relation. [10 points]
- (f) Is there a way to uncover optimal demand using changes in α_j and other terms using the value function? In other words, can you use partial derivatives of $\frac{\partial V}{\partial \alpha_j}$ and other information to uncover y_j^* . Use of the envelope theorem is permitted. [5 points]

1.

(a) convex preference

Show $f(y)$ is $\mathbb{Q}$ -concave.

• $f(y)$ is concave $\Rightarrow f(y)$ is $\mathbb{Q}$ -concave.

Claim) $f(y)$ is concave

i) $2 \ln y_1$ is concave

ii) $3 \ln y_2$ is concave

$\Rightarrow$ The sum of concave functions is concave.

• Concave function is $\mathbb{Q}$ -concave.

• If f is $\mathbb{Q}$ -concave & represents a preference, the preference given by f is convex.

(b) Show the set defined by

$Z := \{ (y, x) \in \mathbb{R}_+^5 \mid y_1 \leq \min \{ 2x_1, 3x_2 \}, y_2 = 3x_2 + x_3 \}$ is convex

(c) Show the set defined by

$B := \{ (x_1, x_2, x_3) \in \mathbb{R}_+^3 \mid p_1 x_1 + p_2 x_2 \leq m \}$
for given p, m is convex.

(d) $\max_{(x_1, x_2, x_3) \in \mathbb{R}_+^3} \underbrace{2 \ln(\min \{ 2x_1, 3x_2 \})}_{=y_1} + 3 \ln(\underbrace{3x_2 + x_3}_{=y_2})$

monotone $\Rightarrow$ binding

$$2\alpha_1, 3\frac{1}{2}\alpha_2, \frac{3}{2}\alpha_3$$

(f) Consider when p_3 is very high and

α_2 now substitutes α_3 .

(Restrictions on parameter values)

(h) y_1, y_2 trade-offs the value

- α_1, α_2 are complementary to produce y_1
- α_2, α_3 are substitutes " y_2
- Common ingredient α_2 links y_1 & y_2
 - purchase of
 - If α_1 or $\alpha_3 \uparrow$, y_1, y_2 should substitute each other
 - A change in α_2